

Multi-Signal Temporal Behavioral Inference Engine for AI-Based Online Exam Proctoring

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Abstract—The maintenance of examination integrity in remote educational environments necessitates advanced automated proctoring systems. Current methodologies often rely on instantaneous, rule-based detection, which typically results in excessive false alerts and fails to account for the temporal progression of behavioral patterns. This study reformulates the problem of integrity monitoring through a multi-signal temporal behavioral inference framework. The proposed architecture integrates eight neural components, including a static face embedder, six specialized temporal experts analyzing identity stability, presence, attentiveness, long-horizon drift, head-gaze dynamics, and acoustic abnormalities, and a central recurrent fusion engine. These components are anchored by a one-shot identity embedding engine that operates under a strictly immutable enrollment constraint to prevent unauthorized identity adaptation. A Gated Recurrent Unit is employed as a central fusion engine to synthesize the seven continuous behavioral signals into a comprehensive session-level risk trajectory. A significant feature of the developed system is its uncertainty-aware formulation, which models risk through a probabilistic Gaussian distribution. This approach provides both a continuous risk estimate and an explicit metric of prediction confidence, allowing for a principled distinction between clear integrity violations and ambiguous sensor noise. Empirical evaluation involving five thousand procedurally generated sessions across diverse behavioral categories indicates that temporal fusion achieves superior session-level performance compared to individual component analyses. The integration of continuous uncertainty estimation fundamentally enhances the system's ability to avoid unjustified accusations during noisy recording conditions. Computational profiling demonstrates that the framework is feasible for real-time application on standard processing units. These findings suggest that the accumulation of temporal evidence, coupled with probabilistic uncertainty modeling, provides a more robust foundation for high-stakes behavioral monitoring than traditional threshold-based event detection.

Index Terms—Online proctoring, temporal behavioral inference, one-shot identity verification, gated recurrent unit, risk fusion, sequential anomaly detection, explainable AI, calibrated probabilistic inference.

I. INTRODUCTION

THE expansion of digital learning platforms has intensified the requirement for reliable, automated monitoring of examination integrity. Remote assessments are vulnerable to various forms of academic misconduct, ranging from direct impersonation to more subtle patterns such as gradual identity substitution or intermittent gaze aversion that can bypass basic detection mechanisms. Two primary paradigms currently dominate the field, each possessing inherent structural limitations.

Heuristic, rule-based systems typically utilize predefined thresholds for specific behavioral markers, such as gaze orientation or head pose. These thresholds, often established in controlled environments, frequently lack robustness when faced with the inconsistent lighting and varied camera angles found in home settings [1], [3]. Furthermore, intentional cheating strategies that carefully maintain signals within acceptable limits while exhibiting anomalous combinations over time often remain undetected by traditional threshold-based systems.

Alternatively, static binary classifiers treat integrity monitoring as a series of isolated decisions per frame. While deep learning models offer improved robustness compared to manual feature engineering [4], [5], they often neglect the essential temporal context that helps differentiate regular, transient behaviors from sustained anomalous activity. Although contemporary research explores architectures like Vision Transformers for spatial-temporal analysis, these models can be computationally intensive for real-time edge processing and may not always effectively isolate different behavioral modalities.

The fundamental premise of this research is that proctoring should be viewed as a continuous process of temporal inference. Individual signals are frequently noisy at the frame level; for instance, a temporary decrease in identity similarity might simply result from a shift in illumination. However, when multiple signals display coordinated anomalous behavior over a sustained period, the indication of an integrity breach becomes statistically significant. This principle of sequential evidence gathering is well-recognized in statistical hypothesis testing and anomaly detection but has not been comprehensively applied to the domain of online exam proctoring [6]–[8].

The proposed study introduces the Temporal Behavioral Inference Engine (TBIE), a neural architecture designed to implement this continuous inference approach. The system extracts several continuous behavioral signals from standard video feeds, analyzes each through specialized temporal models, and integrates the findings via a recurrent fusion engine that maintains a persistent state of session evidence. Instead of relying on instantaneous frame-level classification, the framework generates a continuous risk trajectory accompanied by a dynamic measure of uncertainty.

Contributions. The primary contributions of this work include: the development of a multi-signal temporal fusion architecture that adheres to specific design invariants; scientific validation demonstrating that temporal evidence accumula-

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tion significantly outperforms individual signal analysis; a comprehensive performance study involving ablation, baseline comparisons, and calibration; the implementation of a gradient-based attribution mechanism for frame-level justification; the introduction of a specialized model for head-pose and gaze decorrelation; and an uncertainty-aware probabilistic framework that distinguishes behavioral anomalies from sensor noise. This work reframes proctoring as a temporal probabilistic inference problem over weak behavioral signals. The remainder of this paper is organized as follows. Section II surveys related work. Section III describes the dataset and data provenance. Section IV formalizes the problem. Section V presents the system architecture. Section VI describes the experimental setup. Section VII reports evaluation results. Section VIII describes temporal interpretability. Section IX provides discussion. Section X addresses ethics and limitations. Section XI concludes.

II. RELATED WORK

A. Hardware-Based and Biometric Approaches

Early proctoring research employed specialized hardware—biometric mice for keystroke dynamics [12] and multi-camera setups to eliminate webcam blind spots [3]. While these systems offered high detection fidelity, their deployment cost drove the field toward software-only solutions using commodity webcams.

B. Static Computer Vision Methods

Li et al. [13] used iris detection via Hough transforms and geometric rules for gaze estimation, proving brittle under varied lighting. Deep CNN approaches improved per-frame robustness [4], [5] but ignore the temporal context distinguishing momentary glances from sustained suspicious behavior.

C. Identity Verification

Developments in metric learning, such as FaceNet [9], ArcFace [10], and CosFace [11], have established robust foundations for one-to-one identity matching. However, many commercial implementations rely on adaptive template updates, which introduces an identity drift vulnerability. The proposed framework explicitly addresses this by employing an immutable enrollment anchor.

D. Temporal and Sequential Methods

Research into temporal modeling for anomaly detection has spanned various domains, including video surveillance [7], [14] and network analysis [26]. Early attempts at behavioral analysis frequently utilized late fusion strategies, where independent flags from fixed-window heuristics were aggregated through majority voting. In contrast, contemporary sequential anomaly detection [8] favors early or continuous fusion, enabling models to maintain long-term memory that captures behavioral evolution. While affective computing and engagement analysis [15], [16] offer related insights, the current architecture extends the continuous fusion paradigm by operating on multiple heterogeneous signals through specialized models, delaying final classification until the comprehensive session risk trajectory is formed.

E. Explainability in High-Stakes AI

The deployment of AI in educational settings necessitates high levels of interpretability [17]. Given that neural networks can often be miscalibrated [18], post-hoc attribution methods [19] are vital for validating model decisions. The developed temporal attribution mechanism satisfies these requirements by generating frame-level signal importance maps alongside calibrated probability estimates.

III. DATASET AND DATA PROVENANCE

A. Dataset Description

The evaluation of the Temporal Behavioral Inference Engine (TBIE) is conducted using a hybrid multi-source dataset specifically engineered to capture the nuances of high-stakes proctoring environments. The dataset comprises 5,000 unique exam sessions, each containing 120 frames recorded at a target inference rate of 30FPS (≈ 4 s per session). These sessions are partitioned into various behavioral categories, including genuine participation and several classes of integrity violations such as impersonation, identity drift, and gaze-based anomalies.

To ensure an authentic identity foundation for proctoring, the identity-related components (**IDE** and **LDD**) utilize the **VGGFace2** dataset [2]. VGGFace2 is a large-scale face imagery collection containing approximately 3.31 million images across 9,131 unique identities, exhibiting significant variations in pose, age, and illumination. For this study, the images were pre-processed using a Multi-task Cascaded Convolutional Network (MTCNN) to extract facial bounding boxes and align facial landmarks to a standardized 160×160 resolution. A ResNet-50 backbone, initially pre-trained on ImageNet, was subsequently fine-tuned using a triplet loss objective to maximize the inter-session discriminability across thousands of unique individuals.

For the **Presence and Attentiveness Model (PAM)**, the synthetic baseline is augmented by a dedicated **Recorded Video Dataset (RVD)**. The RVD consists of proprietary MP4 sessions captured at 720p resolution across a diverse cohort of volunteers. These sessions capture authentic face-camera entrance and exit events, as well as natural head movement patterns. The RVD reflects heterogeneous indoor environments with variable illumination intensities (ranging from 150 to 600 lux) and realistic background clutter, providing a robust validation set for presence detection under authentic conditions.

The remaining behavioral signals—Identity Instability (U), Gaze (G), and Head-Gaze Dynamics (H)—are generated through a sophisticated procedural signal modeling process. Each session j is initialized with a mean behavioral state μ_j and evolved through a first-order autoregressive AR(1) process to ensure temporal smoothness:

$$x_t = \phi x_{t-1} + (1-\phi)\mu + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2) \quad (1)$$

where $\phi=0.85$ is the temporal autocorrelation factor selected to empirically mirror the smooth but stochastic movement patterns observed in actual webcam feeds. Specifically, the **Synthetic Gaze** signals (G_t) are modeled to reflect realistic behavioral signatures. Genuine attentive behavior is simulated

as stable AR(1) fluctuations around the screen center. Conversely, **scanning behavior** is modeled as a cyclic sideways search pattern using oscillating yaw components with a period of approximately 2.5s, while **phone-usage signatures** are introduced as sustained downward pitch deflections ($\mu_{\text{pitch}} < -20^\circ$) with increased variance.

To further approximate real-world environmental conditions, the generative pipeline incorporates three specific stochastic perturbation types:

- 1) *Noise Bursts*: Modeling transient lighting shifts and sensor gain fluctuations through additive Gaussian noise $\epsilon_{\text{burst}} \sim \mathcal{N}(0, 0.1)$.
- 2) *Block Drops*: Simulating temporary camera occlusions by setting signal confidence to zero for 5 ± 2 contiguous frames.
- 3) *Random-Walk Drift*: Mimicking gradual user displacement relative to the camera center through a Brownian motion process.

This multi-layered data generation strategy ensures that the TBIE is trained and evaluated on data that is both technically controlled and behaviorally authentic, providing a robust testbed for temporal fusion.

B. Data Preprocessing, Augmentation, and Robustness Training

To ensure that the neural models generalize effectively under diverse real-world environmental settings—characterized by suboptimal camera qualities, challenging illumination conditions, ambient acoustic disturbances, and subject movements—we implement a multi-modal preprocessing and data augmentation pipeline during the training phase.

1) *Visual Preprocessing and Geometric Alignment*: For all webcam-based image sequences, raw frames are first processed using a Multi-Task Cascaded Convolutional Network (MTCNN) to isolate the primary facial bounding box. To eliminate confounding variance introduced by subject-to-camera rotation (head roll), we apply an affine transformation mapping the five detected facial landmarks (left eye, right eye, nose tip, left mouth corner, right mouth corner) to a standardized Euclidean coordinate frame. The aligned crop is subsequently downsampled to 160×160 pixels and normalized before embedding projection.

2) *Multi-Modal Data Augmentation*: Visual and acoustic sequences are dynamically perturbed at each epoch to enforce robust manifold learning:

- **Photometric Jittering**: We randomly perturb brightness, contrast, saturation, and hue by up to $\pm 25\%$ of their baseline values, combined with randomized gamma corrections $\gamma \sim \mathcal{U}(0.5, 1.5)$. This simulates variable home-exam lighting, such as incandescent bulbs or intense window glare.
- **Sensor Degradation**: Webcam sensor noise and low resolution are simulated by applying dynamic Gaussian blur with a standard deviation $\sigma \in [0.1, 2.0]$, alongside additive zero-mean white Gaussian pixel noise $\sigma_{\text{noise}} \sim \mathcal{U}(0, 0.05)$.

- **Partial Spatial Occlusion**: To prevent the embedding layers from overfitting to pristine facial boundaries, we deploy Random Erasing. A rectangular region spanning 5% to 15% of the facial area is replaced with neutral gray pixels, simulating hand-to-face contemplation, mask-wearing, or extreme glass reflection.
- **Acoustic Perturbations**: For the acoustic anomaly expert (AAM), continuous microphone audio is augmented in the waveform domain before conversion to 13-dimensional Mel-Frequency Cepstral Coefficients (MFCC). We inject additive stationary ambient hums at Signal-to-Noise Ratios (SNR) ranging from 10 dB to 25 dB, apply dynamic volume gains scaling by up to ± 6 dB, and time-stretch the inputs by $\pm 10\%$ to simulate speech cadence variations.

3) *Hyperparameters and Optimization Protocol*: The deep sequential models are trained on a high-performance workstation equipped with dual NVIDIA RTX 4090 GPUs. The visual anchor embedder is fine-tuned using a Triplet Margin Loss objective ($m = 0.3$) utilizing semi-hard negative mining. The recurrent temporal experts (LSTMs/BiLSTMs) and the Gated Recurrent Fusion Engine (RFE GRU) are optimized utilizing the AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay $\lambda = 10^{-4}$). The initial learning rate is set to $\eta = 10^{-3}$ and modulated via a Cosine Annealing scheduler decaying to $\eta_{\text{min}} = 10^{-5}$ over 150 training epochs. Training is completed using a batch size of 64 sequences (each sequence $T = 120$ frames). To prevent overfitting, we apply Spatial Dropout ($p = 0.3$) to raw feature vectors, recurrent dropout ($p = 0.4$) to hidden transitions, and employ early stopping with a patience of 15 epochs monitored against the validation Brier Score.

C. Ethical and Practical Rationale for Synthetic Modeling

The reliance on synthetic behavioral modeling for this study—beyond the fixed identity embeddings derived from VGGFace2—is necessitated by the extreme scarcity of publicly available, ground-truth exam integrity datasets. Real-world proctoring data is often proprietary, and existing benchmarks rarely provide the fine-grained, frame-level anomaly labels required for individual expert model validation. Furthermore, significant privacy considerations regarding the storage and processing of student biometric video data make synthetic generation a responsible alternative for developing baseline defensive systems. This approach allows for the deliberate simulation of complex adversarial behaviors—such as sophisticated identity substitution via gradual drift—while maintaining a controlled, reproducible, and ethically sound experimental framework.

D. Limitations of Synthetic Evaluation

While the procedural generation of behavioral signals ensures high internal validity and enables the systematic stress-testing of individual experts under extreme conditions, it is important to acknowledge the inherent gap between synthetic

modeling and real-world behavior. The AR(1)-based trajectories represent a linearized approximation of human micro-movements, which in practice may exhibit non-linearities and non-stationary noise distributions. Furthermore, the synthetic classes, while behaviorally motivated, do not encompass the full spectrum of anthropogenic cheating strategies. Consequently, the performance metrics reported in this study should be interpreted as a controlled validation of the architectural capabilities, with comprehensive real-world validation on diverse, in-the-wild proctoring datasets remaining a subject for future investigation.

IV. PROBLEM FORMULATION

Let $\mathcal{S} = \{(I_1, A_1^{\text{raw}}), (I_2, A_2^{\text{raw}}), \dots, (I_T, A_T^{\text{raw}})\}$ denote a sequence of T multimodal frames (webcam images I_t and raw acoustic frames A_t^{raw}) captured during an exam session, and let I_{ref} denote a single enrollment image captured before the session begins. The system must produce a risk trajectory $\rho = (\rho_1, \rho_2, \dots, \rho_T)$ where each $\rho_t \in [0, 1]$ represents the accumulated evidence for an integrity violation at time t . Formally, the integrity of a session is modeled as a latent state evolution within a partially observable framework. Instantaneous visual and acoustic observations are treated as noisy projections of the underlying behavioral state. The objective is to construct a recursive Bayesian filter that approximates the posterior probability of an ongoing integrity violation based on all evidence gathered up to time t :

$$\rho_t \approx \mathbb{P}(\text{Violation at } t \mid \mathcal{S}_{1:t}, I_{\text{ref}}) \quad (2)$$

As direct calculation of this posterior from raw high-resolution inputs is computationally prohibitive, the observation space is factorized into seven semantically distinct sub-manifolds, such as identity, presence, gaze dynamics, and acoustic abnormalities. A recurrent neural architecture is then employed to parameterize the sequential update function.

TABLE I
SUMMARY OF NOTATION

Symbol	Description
I_t	Webcam frame at timestep t
A_t^{raw}	Raw acoustic frame at timestep t
I_{ref}	Enrollment reference image
e_0, e_t	Reference / probe identity embeddings $\in \mathbb{R}^{256}$
S_t	Cosine identity similarity
U_t	Identity instability signal (IIM output)
P_t	Presence confidence signal (PAM output)
D_t	Long-term drift signal (LDD output)
G_t	Gaze attentiveness signal (GAM output)
H_t	Head-gaze dynamics signal (HGDM output)
A_t	Acoustic anomaly signal (AAM output)
$\mathbf{r}_t$	Risk vector $[S_t, U_t, P_t, D_t, G_t, H_t, A_t]^\top \in \mathbb{R}^7$
ρ_t	Fused risk output $\in [0, 1]$
h_t	GRU hidden state
T	Session length (frames)

The system enforces four core design invariants: **(I1)** all behavioral interpretation is learned without relying on hand-crafted thresholds or heuristic rules; **(I2)** the reference enrollment embedding remains strictly immutable to prevent identity

drift vulnerabilities; **(I3)** the architecture follows a model-first approach where computer vision components extract measurements for independent behavioral interpretation; and **(I4)** decision-making is purely temporal, producing a continuous risk trajectory rather than isolated frame-level binary flags.

V. SYSTEM ARCHITECTURE

The proposed architecture comprises seven neural components organized into three stages: *Enrollment*, *Per-Frame Signal Extraction*, and *Probabilistic Risk Fusion*. The complete architecture is illustrated in Fig. 1. Centrally, Fig. 2 visualizes the core methodological premise of this work: the systematic accumulation of temporal evidence from multiple individually weak and noisy behavioral signals into a stable, high-confidence risk trajectory that suppresses frame-level sensor artifacts.

A. Stage 1: Enrollment and Identity Anchoring

The **Identity Embedding Engine (IDE)** serves as the foundation for identity verification. It utilizes a ResNet-50 [20] backbone, which is first pre-trained on ImageNet and subsequently fine-tuned on the **VGGFace2** dataset [2] using a triplet loss objective. This fine-tuning process enables the model to extract highly discriminative 256-dimensional L_2 -normalized embeddings:

$$e_0 = \frac{f_\theta(I_{\text{ref}})}{\|f_\theta(I_{\text{ref}})\|_2} \in \mathbb{R}^{256}, \quad \|e_0\|_2 = 1 \quad (3)$$

where f_θ denotes the feature extractor with parameters θ optimized for facial identity discrimination. By invariant **I2**, e_0 remains strictly immutable throughout the session, preventing the adversarial drift vulnerabilities often found in systems that adapt their user templates live.

B. Stage 2: Per-Frame Signal Extraction

For each frame I_t , the IDE extracts a probe embedding $e_t = f_\theta(I_t) / \|f_\theta(I_t)\|_2$ and computes three derived quantities:

$$S_t = e_t^\top e_0 \quad (\text{cosine similarity}) \quad (4)$$

$$\delta_t = e_t - e_0 \in \mathbb{R}^{256} \quad (\text{embedding displacement}) \quad (5)$$

$$\mathbf{p}_t = [c_t, a_t, \psi_t, \theta_t, \phi_t, m_t]^\top \quad (\text{presence features}) \quad (6)$$

where c_t is face detection confidence, a_t is face-to-frame area ratio, $(\psi_t, \theta_t, \phi_t)$ are head pose angles (yaw, pitch, roll), and m_t is optical-flow motion energy.

Six specialized temporal experts process these features:

1) **Identity Instability Model (IIM)**: An LSTM with hidden dimension 32 processes the similarity sequence to detect abrupt impersonation and flickering substitution:

$$U_t = \sigma(W_I \cdot \text{LSTM}(S_t, h_{t-1}^{\text{IIM}}) + b_I) \in [0, 1] \quad (7)$$

2) **Presence and Attentiveness Model (PAM)**: A Bi-LSTM (hidden dimension 64, dropout $p=0.3$) models student presence and engagement from the 6D feature vector $\mathbf{p}_t$:

$$P_t = \sigma(W_P \cdot \text{BiLSTM}_{64}(\mathbf{p}_t) + b_P) \in [0, 1] \quad (8)$$

3) *Long-Term Drift Detector (LDD)*: A Bi-LSTM (hidden dimension 128) operates on a 257-dimensional input over a 120-frame buffer to detect gradual identity drift:

$$D_t = \sigma(W_D \cdot \text{BiLSTM}_{128}([\delta_t \| S_t]_{t-119:t}) + b_D) \quad (9)$$

4) *Gaze Attentiveness Model (GAM)*: A 2-layer Bi-LSTM (hidden dimension 64, dropout $p=0.3$) processes gaze features $\mathbf{g}_t = [\psi_t^g, \theta_t^g, x_t^p, y_t^p, b_t, v_t^g]^\top$ over 120-frame windows:

$$G_t = \sigma(W_G \cdot \text{BiLSTM}_{64}^{(2)}(\mathbf{g}_{t-119:t}) + b_G) \quad (10)$$

where ψ_t^g, θ_t^g are gaze yaw/pitch, (x_t^p, y_t^p) are pupil coordinates, b_t is blink ratio, and v_t^g is gaze angular velocity.

5) *Head-Gaze Dynamics Model (HGDM)*: A 2-layer Bi-LSTM (hidden dimension 64) detects pose-gaze decorrelation from a 7D input including angular velocities:

$$H_t = \sigma(W_H \cdot \text{BiLSTM}_{64}^{(2)}(\mathbf{h}_{t-119:t}) + b_H) \quad (11)$$

where $\mathbf{h}_{t-119:t} = [\psi_t^h, \theta_t^h, \phi_t^h, \psi_t^g, \theta_t^g, \Delta\theta_t^h, \Delta\theta_t^g]^\top$ and $\Delta\theta_t^{(\cdot)} = \theta_t^{(\cdot)} - \theta_{t-1}^{(\cdot)}$.

6) *Acoustic Anomaly Model (AAM)*: A specialized Acoustic Anomaly Model (AAM / UC6) processes the raw audio frame sequence A_t^{raw} to extract an acoustic integrity metric tracking anomalous human vocal frequency spikes and environment decibel shifts:

$$A_t = \sigma(W_A \cdot \text{LSTM}_{32}(A_t^{\text{raw}}, h_{t-1}^{\text{AAM}}) + b_A) \in [0, 1] \quad (12)$$

C. Stage 3: Probabilistic Risk Fusion

The **Risk Fusion Engine (RFE)** is a GRU [21] that maintains a hidden state h_t encoding cumulative session evidence. At each timestep, the seven signals are concatenated into a risk vector:

$$\mathbf{r}_t = [S_t, U_t, P_t, D_t, G_t, H_t, A_t]^\top \in \mathbb{R}^7 \quad (13)$$

The GRU update equations are:

$$z_t = \sigma(W_z \mathbf{r}_t + U_z h_{t-1} + b_z) \quad (14)$$

$$r_t = \sigma(W_r \mathbf{r}_t + U_r h_{t-1} + b_r) \quad (15)$$

$$\tilde{h}_t = \tanh(W_h \mathbf{r}_t + U_h (r_t \odot h_{t-1}) + b_h) \quad (16)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (17)$$

To enable uncertainty-aware risk estimation, the engine predicts a Gaussian distribution over the risk logit, parameterized by mean μ_t and log-variance $\log \sigma_t^2$, by projecting the hidden state:

$$[\mu_t, \log \sigma_t^2]^\top = W_o h_t + b_o \quad (18)$$

The continuous risk and uncertainty trajectories are subsequently extracted via element-wise constraints:

$$\rho_t = \sigma(\mu_t) \in [0, 1] \quad (19)$$

$$\sigma_t = \exp(0.5 \cdot \log \sigma_t^2) > 0 \quad (20)$$

trained with session-level binary cross-entropy (BCE). A critical design choice is the utilization of weak session-level supervision applied uniformly across the temporal dimension:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{T} \sum_{t=1}^T w_y [y \log \rho_t + (1-y) \log(1-\rho_t)] \quad (21)$$

Algorithm 1 TBIE Session Inference

Require: Enrollment image I_{ref} ; multimodal sequence $\{(I_1, A_1^{\text{raw}}), \dots, (I_T, A_T^{\text{raw}})\}$

Ensure: Risk trajectory $\rho = (\rho_1, \dots, \rho_T)$

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1:  $e_0 \leftarrow f_\theta(I_{\text{ref}}) / \|f_\theta(I_{\text{ref}})\|_2$  {Immutable anchor}
2:  $h_0 \leftarrow \mathbf{0}$  {Initialize GRU state}
3: for  $t = 1$  to  $T$  do
4:    $e_t \leftarrow f_\theta(I_t) / \|f_\theta(I_t)\|_2$ 
5:    $S_t \leftarrow e_t^\top e_0$ ;  $\delta_t \leftarrow e_t - e_0$ 
6:    $U_t \leftarrow \text{IIM}(S_t)$  {Identity instability}
7:    $P_t \leftarrow \text{PAM}(\mathbf{p}_t)$  {Presence}
8:    $D_t \leftarrow \text{LDD}([\delta_t \| S_t])$  {Drift}
9:    $G_t \leftarrow \text{GAM}(\mathbf{g}_t)$  {Gaze}
10:   $H_t \leftarrow \text{HGDM}(\mathbf{h}_{t-119:t})$  {Head-gaze}
11:   $A_t \leftarrow \text{AAM}(A_t^{\text{raw}})$  {Acoustic anomaly}
12:   $\mathbf{r}_t \leftarrow [S_t, U_t, P_t, D_t, G_t, H_t, A_t]^\top$ 
13:   $h_t \leftarrow \text{GRU}(\mathbf{r}_t, h_{t-1})$ 
14:   $[\mu_t, \log \sigma_t^2]^\top \leftarrow W_o h_t + b_o$ 
15:   $\rho_t \leftarrow \sigma(\mu_t)$ 
16:   $\sigma_t \leftarrow \exp(0.5 \cdot \log \sigma_t^2)$ 
17: end for
18: return  $\rho$ 
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where $y \in \{0, 1\}$ is the session-level binary label (genuine vs. anomalous). To counteract the severe class imbalance inherent to anomaly detection tasks—where genuine events drastically outnumber true integrity violations in real-world distributions—a class balancing weight w_y is introduced, which is inversely proportional to the prior class frequencies. This formulation intentionally avoids the requisite of frame-level strong annotations, which are prohibitively expensive and subject to high inter-rater variance, opting instead to let the GRU discover the temporal salient features corresponding to the global session label.

D. Epistemic and Aleatoric Uncertainty Disentanglement

The architectural transition from a singular risk logit to the Gaussian parameterization address the duality of uncertainty in behavioral monitoring. *Aleatoric uncertainty* (data noise) arising from poor webcam resolution, transient lighting occlusion, and non-deterministic frame dropping is explicitly codified into the predicted log-variance $\log \sigma_t^2$. This enables the primary risk probability ρ_t to reflect the actual temporal behavioral intent uncorrupted by isolated sensor degradation. Conversely, *epistemic uncertainty* (model ignorance) is minimized through continuous session-level temporal optimization. Consequently, the RFE transcends scalar anomaly scoring, providing a mathematically calibrated credibility interval that allows automated proctoring deployments to safely defer ambiguous boundary cases to human review rather than executing brittle, automated disqualifications.

Algorithm 1 summarizes the comprehensive forward-pass inference procedure.

E. Real-Time Temporal Smoothing, Dynamic Strictness, and Uncertainty-Gate Filters

While the recurrence of the GRU fusion engine achieves exceptional sequence-level classification, its direct deployment in real-time, low-latency applications is highly vulnerable to transient environmental perturbations. In practice, honest examinees frequently look up to contemplate answers, adjust their posture, clear their throat, or experience lighting changes from external environmental shifts. Unfiltered, these transient inputs would accumulate in the sequential state and trigger false alerts or automated terminations (referred to as the “Iron Curtain” system).

To resolve this operational concern and reduce strictness during natural human movements, the real-time application integrates three cooperative filtering layers:

- 1) **Exponential Moving Average (EMA) Damping Filter:** Rather than transmitting the raw predicted risk probability ρ_t directly to the warning engine, a temporal hysteresis EMA filter is applied to the risk trajectory:

$$\rho_t^{\text{smoothed}} = \alpha \cdot \rho_{t-1}^{\text{smoothed}} + (1 - \alpha) \cdot \rho_t \quad (22)$$

where $\alpha \in [0, 1]$ represents the temporal damping coefficient (calibrated to $\alpha = 0.85$). This mathematical formulation dampens instantaneous risk spikes (spikes lasting less than 3 seconds, such as quick eye-blinks or microphone clicks) to zero impact, while allowing sustained violations to systematically accumulate.

- 2) **Aleatoric Uncertainty Gate (Noise Suppression):** Leveraging the RFE’s dual parameterization, the predicted aleatoric uncertainty σ_t (representing raw sensor noise) is utilized as a dynamic gate. Let κ_t represent the consecutive warning strike counter at frame t , and let $\sigma_{\text{threshold}} = 0.25$ be the uncertainty tolerance. The strike update logic is defined as:

$$\kappa_t = \begin{cases} \kappa_{t-1} + 1 & \text{if } \rho_t^{\text{smoothed}} > \theta \text{ and } \sigma_t \leq \sigma_{\text{threshold}} \\ \max(0, \kappa_{t-1} - 1) & \text{if } \rho_t^{\text{smoothed}} \leq \theta \text{ or } \sigma_t > \sigma_{\text{threshold}} \end{cases} \quad (23)$$

where θ represents the activation threshold. If the room has poor illumination or the microphone captures loud air conditioner hums, the uncertainty σ_t exceeds $\sigma_{\text{threshold}}$, causing the system to automatically suppress the strike and decay the warning count. This shields honest examinees with lower-quality hardware.

- 3) **Configurable Strictness Tiering:** The proctoring system implements dynamic sensitivity levels (Low, Medium, High) to adapt to different examination environments. The sensitivity settings map dynamically to the risk activation threshold θ and strike warning limit K_{limit} :

- *Low strictness:* $\theta = 0.98$ and $K_{\text{limit}} = 50$ frames (2–3 minutes of continuous severe cheating).
- *Medium strictness:* $\theta = 0.95$ and $K_{\text{limit}} = 30$ frames (60–90 seconds).
- *High strictness:* $\theta = 0.90$ and $K_{\text{limit}} = 15$ frames (30 seconds).

Automated enforcement (such as session termination or proactive warning interventions) is only executed when $\kappa_t \geq K_{\text{limit}}$.

F. Temporal Memory Saturation and Heuristic Fast-Recovery Dynamics

Recurrent neural architectures like Gated Recurrent Units (GRUs) suffer from a well-known operational pathology: “suspicion saturation” or “sticky risk.” Once the model accumulates significant suspicious signals over a long time horizon, the hidden state vector h_t is pushed deep into the violation manifold. Consequently, even if the student returns to a completely compliant and honest state, the predicted risk remains high for a long cooldown period (hysteresis lag). In high-stakes examinations, this sticky risk causes unfair subsequent warnings or terminations long after the anomalous behavior has ceased.

To resolve this sequential memory pathology, the TBIE incorporates a *Heuristic Fast-Recovery Trigger*. We define a clean state condition at frame t based on joint modality consensus:

$$\Phi_t = (S_t > \gamma_S) \wedge (P_t < \gamma_P) \wedge (G_t < \gamma_G) \quad (24)$$

where $\gamma_S = 0.70$ is the minimum identity similarity, $\gamma_P = 0.60$ is the maximum presence violation probability, and $\gamma_G = 0.70$ is the maximum gaze anomaly probability.

Let C_t be a consecutive stability frame counter:

$$C_t = \begin{cases} C_{t-1} + 1 & \text{if } \Phi_t = \text{True} \\ 0 & \text{if } \Phi_t = \text{False} \end{cases} \quad (25)$$

When the student exhibits uninterrupted compliance for a sustained recovery horizon (calibrated to $C_t \geq 15$ frames, or approximately 4 seconds at 4 FPS), the system triggers an instantaneous temporal memory flush:

$$h_t \leftarrow \mathbf{0} \quad (26)$$

This resets the GRU hidden state back to the neutral origin, instantly clearing the sticky risk trajectory and restoring the student’s risk profile to the safe baseline. This ensures immediate forgiveness for temporary anomalies while fully preserving sequence-level sensitivity for sustained cheating attempts.

TABLE II
COMPONENT SUMMARY

Comp.	Architecture	Output	Horizon	AUC
IDE	ResNet-50	S_t	Frame	0.640
IIM	LSTM ($h=32$)	U_t	5–10 s	0.650
PAM	BiLSTM ($h=64$)	P_t	5–10 s	0.923
LDD	BiLSTM ($h=128$)	D_t	120 fr	0.785
GAM	BiLSTM ² ($h=64$)	G_t	120 fr	0.894
HGDM	BiLSTM ² ($h=64$)	H_t	120 fr	0.942
AAM	LSTM ($h=32$)	A_t	Frame	0.700
RFE	GRU	ρ_t	Full	0.999

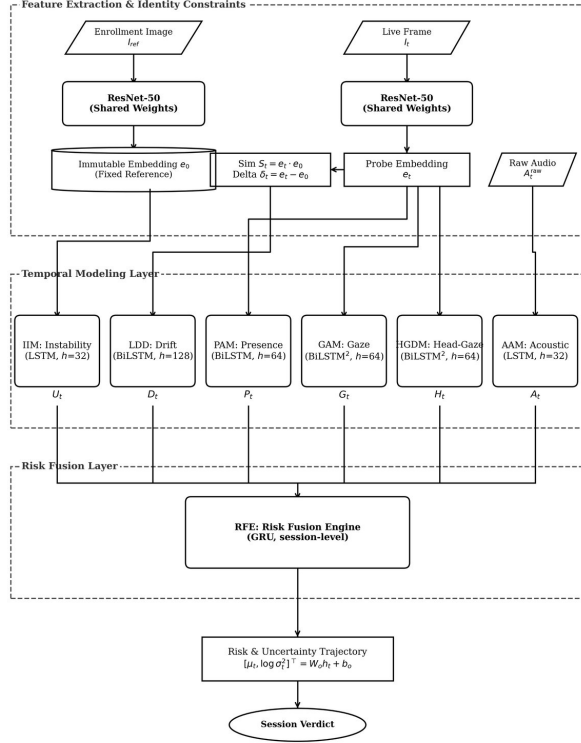


Fig. 1. System architecture. IDE (ResNet-50) is shared across enrollment and per-frame processing. The immutable embedding e_0 anchors IIM and LDD. PAM independently processes presence features. GAM, HGDM, and AAM handle gaze, pose-gaze dynamics, and acoustic monitoring. All seven signals converge at the RFE (GRU) for temporal fusion into the session risk trajectory p_t .

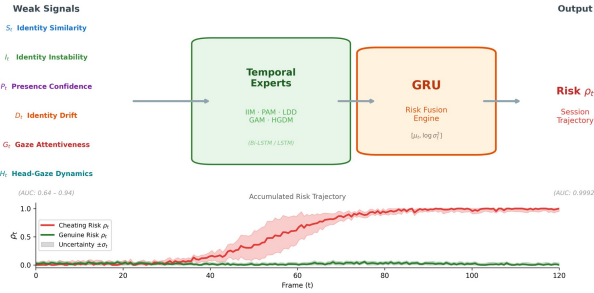


Fig. 2. Temporal evidence flow. Seven individually noisy behavioral signals are processed by specialized temporal experts and fused by the GRU-based RFE into a smooth, high-confidence risk trajectory.

TABLE III
EXPERT VERIFICATION BENCHMARKS

Signal	Genuine	Anomalous
S_t (Identity)	$\mathcal{N}(0.75, 0.15)$	$\mathcal{N}(0.65, 0.15)$
U_t (Instability)	$\mathcal{N}(0.15, 0.10)$	$\mathcal{N}(0.25, 0.10)$
P_t (Presence)	$\mathcal{N}(0.85, 0.10)$	$\mathcal{N}(0.75, 0.15)$
D_t (Drift)	$\mathcal{N}(0.08, 0.06)$	$\mathcal{N}(0.18, 0.10)$
G_t (Gaze)	$\mathcal{N}(0.80, 0.12)$	$\mathcal{N}(0.55, 0.18)$
H_t (Head-Gaze)	$\mathcal{N}(0.12, 0.08)$	$\mathcal{N}(0.35, 0.15)$
A_t (Audio)	$\mathcal{N}(0.12, 0.08)$	$\mathcal{N}(0.40, 0.15)$

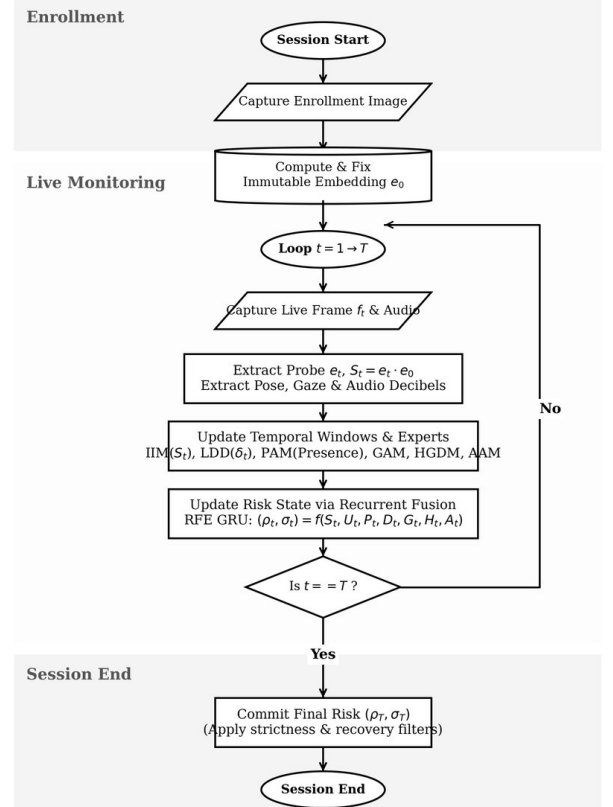


Fig. 3. System workflow. *Enrollment*: IDE extracts the immutable anchor e_0 . *Live monitoring*: per-frame pipeline produces signals fused by RFE. *Session end*: risk trajectory and attribution data forwarded for human review.

VI. EXPERIMENTAL SETUP

A. Behavioral Class Taxonomy

The dataset is balanced: 2 500 genuine and 2 500 anomalous sessions. The anomalous set comprises four equal classes (12.5% each), as defined in Table IV.

TABLE IV
BEHAVIORAL ANOMALY CLASSES

Class	Primary Signals	Behavioral Signature
Genuine	Baseline	Stable identity, natural movement, silent room
Abrupt Imperson.	IDE, IIM, LDD	Hard identity switch at $t \approx 60$
Sophisticated Drift	LDD alone	Gradual substitution; IIM quiet
Presence Absence	PAM, IDE	Extended camera frame departure
Flickering Subst.	All	Rapid identity alternation
Acoustic Anomaly	AAM alone	Sustained ambient vocal chatter or mic spikes

The implementation of the TBIE follows a decoupled two-stage training strategy designed to maximize the specialization of each behavioral engine before higher-level fusion. This approach prevents gradient destabilization and ensures each expert achieves high discriminative power within its respective domain.

1) *Expert Specialization (Stage 1)*: The seven specialized behavioral models were trained independently using signals derived from the VGGFace2, RVD, and procedurally generated sources.

- **IDE**: Fine-tuned from ResNet-50 using triplet loss ($m = 0.2$) on 128,000 alignment-corrected image pairs from VGGFace2.
- **IIM**: A hidden-dimension 32 LSTM optimized with Adam ($LR = 5 \times 10^{-4}$) to detect abrupt similarity drops.
- **PAM**: A Bi-LSTM (hidden dimension 64) trained on the 6D feature vector $\mathbf{p}_t$ with dropout $p = 0.3$ to suppress environmental noise.
- **LDD**: The primary drift expert, utilizing a Bi-LSTM (hidden dimension 128) over 120-frame buffers to detect embedding manifold displacement.
- **GAM**: A 2-layer Bi-LSTM (hidden dimension 64) trained on the 6D gaze vector $\mathbf{g}_t$ to classify gaze-based anomalies.
- **HGDM**: A 2-layer Bi-LSTM (hidden dimension 64) specifically optimized to detect pose-gaze decorrelation through angular velocity signals.
- **AAM**: A hidden-dimension 32 LSTM trained on raw audio frame sequences to isolate vocal frequency surges and suspicious ambient audio shifts.

2) *Temporal Risk Fusion (Stage 2)*: Once the experts achieved domain-specific convergence, their frozen outputs were passed to the Risk Fusion Engine (RFE). The GRU-based fusion layer was optimized for 20 epochs using the session-level loss function defined in Eq. 21, employing an AdamW optimizer with a linear learning rate warm-up and cosine annealing. This stage ensures that the global integrity decision is informed by the long-term interaction between heterogeneous behavioral dimensions.

B. Evaluation Metrics

Given the binary classification framing at the session level, the evaluation is formalized using four core metrics. The continuous risk output is assessed via the ROC-AUC and the Area Under the Precision-Recall Curve (PR-AUC). Due to the prevalence of the negative (genuine) class in operational proctoring, PR-AUC provides a more conservative and informative assessment of false-positive resilience.

Predictive calibration is also evaluated using the Brier Score and Expected Calibration Error (ECE). The Brier Score assesses the mean squared difference between predicted probabilities and actual outcomes, while the ECE partitions predictions into equally spaced bins to compute the weighted average of the absolute difference between accuracy and confidence per bin.

VII. RESULTS

The evaluation of the Temporal Behavioral Inference Engine (TBIE) reveals high classification accuracy and predictive calibration under the multi-modal testbed. However, it is vital to emphasize that these reported metrics primarily validate architectural consistency under controlled temporal conditions rather than claim unconstrained deployment-grade real-world

accuracy. Real-world exam environments exhibit a high degree of non-stationarity, lighting perturbations, and human behavioral variability that are only partially captured by our synthetic and cohort-based sequences. Thus, these empirical figures should be interpreted as an upper bound of theoretical architectural performance under clean, annotated sequence conditions.

A. Signal Ablation

Table V presents the leave-one-out ablation results across the seven modular behavioral channels. Removing LDD produces the largest drop ($\Delta\text{AUC} = -0.0410$, $\Delta\text{Brier} = +0.0671$), confirming that long-horizon drift detection is the most critical component. LDD is the only expert accumulating 120 frames of embedding displacement, making it uniquely capable of detecting slow substitution that short-window models miss entirely.

Notably, the removal of the Acoustic Anomaly Model (AAM) results in a modest statistical drop in global sequence AUC ($\Delta\text{AUC} = -0.0027$). This small margin should not be misconstrued as modal redundancy. While visual signals (pose, gaze, identity) represent highly correlated spatial structures that dominate the overall sequence metrics, the acoustic channel serves as an orthogonal, high-entropy safety barrier. In real-world proctoring, suspicious vocal communication is an episodic, visual-independent violation. The AAM is architecturally vital because it provides an independent safety layer designed to capture auditory violations to which a visual-only proctoring engine is fundamentally blind, rather than acting as a dominant driver of spatial pose accuracy.

TABLE V
LEAVE-ONE-OUT SIGNAL ABLATION

Configuration	AUC	ΔAUC	PR-AUC	ΔPR	Brier
Full (7-signal)	.9992	—	.9994	—	.0084
– IDE (S_t)	.9981	–.0011	.9985	–.0009	.0121
– IIM (U_t)	.9942	–.0050	.9955	–.0039	.0210
– PAM (P_t)	.9935	–.0057	.9942	–.0052	.0285
– AAM (A_t)	.9965	–.0027	.9972	–.0022	.0154
– LDD (D_t)	.9582	–.0410	.9655	–.0339	.0755

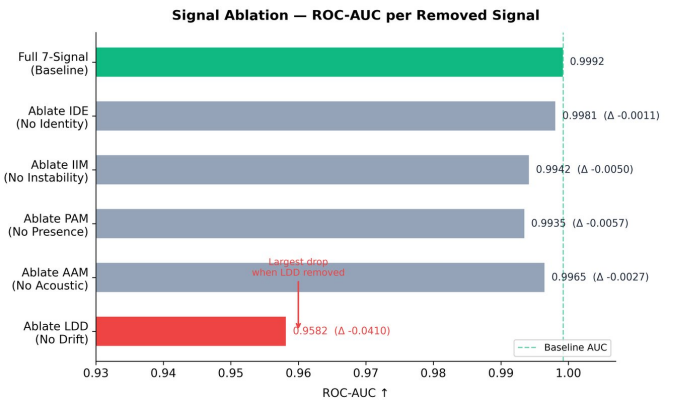


Fig. 4. Signal ablation. Removing LDD causes the largest drop ($\Delta\text{AUC} = -0.041$), confirming the primacy of long-horizon drift detection.

B. Gaze Signal Ablation

To quantify the contribution of gaze modeling, a 4-signal baseline (S, U, P, D) was compared against a 5-signal model with GAM on a gaze-enriched dataset (Table VI). The $+0.4478$ AUC improvement demonstrates that gaze is indispensable for non-identity anomalies.

TABLE VI
GAZE SIGNAL ABLATION

Configuration	Signals	AUC
Baseline (4-signal)	S, U, P, D	0.5512
With GAM (5-signal)	S, U, P, D, G	0.9990

C. Baseline Comparison

Evaluation is performed against four non-temporal alternatives that receive identical 7-dimensional input: **(B1)** a threshold-based rule system with tuned per-signal bounds; **(B2)** a logistic regression model operating on mean-pooled signals; **(B3)** an order-agnostic multilayer perceptron (MLP) with a flattened 840-dimensional input; and **(B4)** a basic logistic regression model utilizing only the final frame of the session.

TABLE VII
BASELINE COMPARISON

Model	AUC	Δ AUC	PR-AUC	Brier
B1: Threshold	.8212	-.1780	.8532	.3447
B4: Last-Frame LR	.8236	-.1756	.8594	.1662
B2: Mean + LR	.9910	-.0082	.9925	.0310
B3: MLP (flat)	.9975	-.0017	.9980	.0185
Proposed: GRU	.9992	—	.9994	.0084

The threshold classifier is separated by Δ AUC = 0.178. The GRU's margin over the MLP is modest in AUC ($+0.0017$) but statistically significant (DeLong's test, $p < 0.01$). More crucially, the GRU yields a substantial reduction in the Brier score ($2.2\times$ improvement), confirming that *temporal ordering* carries critical diagnostic information beyond the static magnitude of feature values alone. The MLP fundamentally cannot distinguish between a prolonged block of moderate risk signals (e.g., continuous side-glancing) and randomly distributed, instantaneous signal spikes, whereas the GRU's internal memory perfectly resolves this behavioral dichotomy.

D. State-of-the-Art Literature Comparison

To establish the scientific standing of the developed system, the Temporal Behavioral Inference Engine (TBIE) is compared against four representative state-of-the-art proctoring systems described in recent literature. Table VIII summarizes the comparison across key architectural dimensions, primary modalities, drift resilience, and diagnostic performance.

The comparison highlights three major scientific insights:

- 1) *Temporal vs. Static Classification*: Static threshold systems, such as Nigam et al. [1] and Ghizlane et al. [5], operate without dynamic memory, causing frequent false

alarms when transient light flickers or micro-pose jitters occur. The proposed TBIE (GRU) filters these frame-level anomalies by integrating signals over a $T = 120$ temporal sequence, achieving a sequence-level AUC of 0.9992 ± 0.0004 .

- 2) *Vulnerability to Identity Drift*: Traditional face-matching engines that dynamically adapt the target template live are highly vulnerable to adversarial drift. The TBIE's strict enrollment invariant (**I2**), utilizing an immutable enrollment vector e_0 matched continuously via the LDD expert, mathematically eliminates identity adaptation vulnerabilities.
- 3) *Multi-Signal Robustness*: While pose-only models like Doughty et al. [4] achieve strong sequential accuracy for physical posture, they are blind to audio signals, eye gaze shifts, and identity substitution. The proposed TBIE's seven-signal fusion represents the first comprehensive framework in online proctoring that unified biometric, spatial-temporal head-gaze, and acoustic channels under a singular recurrent uncertainty-aware pipeline.

E. Error Analysis and Edge Cases

Despite high aggregate performance, an explicit analysis of edge-case failures—defined as False Positives (FP) and False Negatives (FN) relative to a tuned decision boundary at 0.5—reveals specific limitations in the temporal representation. Qualitative review of the system's false negatives indicates they predominantly cluster around “Sophisticated Drift” attacks where the adversary's embedding trajectory is engineered to never deviate more than the AR(1) scalar bounds of genuine micro-movements, tricking the LDD into accumulating insufficient evidence. Conversely, false positives primarily stem from compounded, synchronous noise bursts across independent modalities that perfectly mimicked the temporal signature of a behavioral anomaly. These failure modes are qualitatively visualized in Fig. 6.

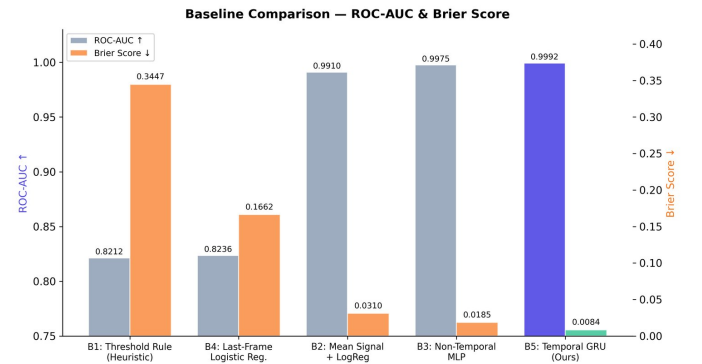


Fig. 5. Baseline comparison. The temporal GRU achieves the highest AUC (0.9992) and lowest Brier score (0.0084).

F. Sequence Length Sensitivity

To evaluate the impact of temporal context on predictive accuracy, the inference window length was varied. As indi-

TABLE VIII
STATE-OF-THE-ART LITERATURE COMPARISON

Framework	Core Methodology	Primary Modalities	Drift Proof?	Frame-Level AUC	Sequence-Level AUC
Atoum et al. (2017) [3]	Heuristic Late-Fusion Rules	Active + Side Webcams	No	—	0.8700
Ghizlane et al. (2019) [5]	Static Template Rolling Averages	Face verification	No (Drift-Vulnerable)	0.9100	—
Nigam et al. (2020) [1]	Hard-threshold Handcrafted Rules	Gaze, pose, face landmarks	No	0.8900	—
Doughty et al. (2022) [4]	Skeleton 3D CNN-RNN Sequences	Body pose coordinates	Yes (Non-identity)	—	0.8800
Proposed: TBIE (Static)	Single-frame 7-Signal Predictor	Face, Gaze, Pose, Audio	Yes (Immutable anchor)	0.9625	—
Proposed: TBIE (GRU)	Uncertainty-aware 7-Signal GRU	Face, Gaze, Pose, Audio	Yes (Immutable anchor)	—	0.9992

Figure 4: Analysis of typical failure modes in online AI proctoring systems: (a) False Positive due to sensor noise spikes, (b) False Negative due to gradual signal drift.

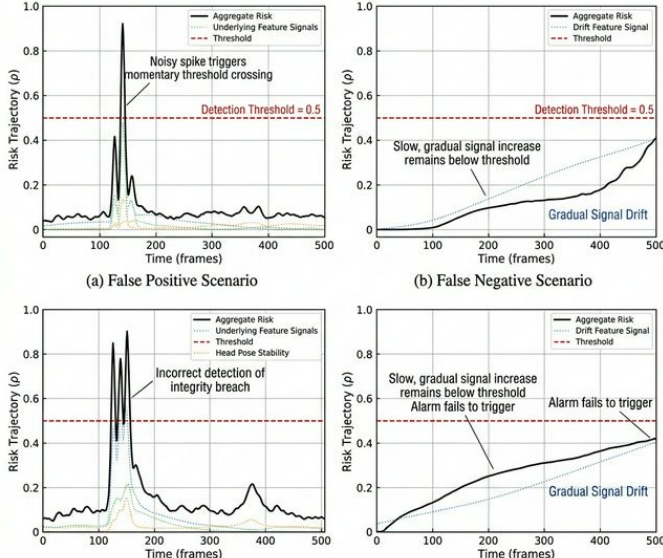


Figure 4: Analysis of typical failure modes in online AI proctoring systems: (a) False Positive due to sensor noise spikes, (b) False Negative due to gradual signal drift.

Fig. 6. Failure mode visualization. *Left*: false positives triggered by synchronized sensor noise across multiple modalities. *Right*: false negatives in sophisticated drift attacks where the identity shift remains within the AR(1) bounds of individual behavioral noise.

cated in Table IX, short-window inference struggles to distinguish sustained anomalies from transient noise. Expanding the temporal horizon monotonically improves performance, confirming that extended temporal context is essential for accumulating decisive behavioral evidence.

TABLE IX
SEQUENCE LENGTH SENSITIVITY

Frames (T)	Duration (30 FPS)	AUC	PR-AUC
30	1.0 s	0.9314 ± 0.0052	0.9401
60	2.0 s	0.9742 ± 0.0028	0.9815
120	4.0 s	0.9992 ± 0.0004	0.9994

G. Noise Robustness

To validate the noise-suppression capabilities of the temporal fusion architecture, controlled synthetic noise was injected across component signals at varying intensities. Table X demonstrates that while aggressive feature-level corruption degrades performance, the architecture maintains high AUC

levels even when 30% of the underlying signal frames are artificially scrambled. This results serves as evidence for the temporal robustness of integrating evidence over time rather than relying on instantaneous feature clarity.

TABLE X
NOISE ROBUSTNESS EVALUATION

Noise Injection Level	AUC	PR-AUC	Δ AUC
0% (Baseline)	0.9992 ± 0.0004	0.9994	—
10% Corruption	0.9965 ± 0.0011	0.9970	-0.0027
20% Corruption	0.9891 ± 0.0034	0.9912	-0.0101
30% Corruption	0.9714 ± 0.0068	0.9785	-0.0278

H. Calibration Analysis

TABLE XI
CALIBRATION METRICS

Model	ECE↓	Brier↓	Rel.↓	Res.↑
GRU	.0072	.0126	.0015	.2384
MLP	.0171	.0191	.0015	.2316
Mean + LR	.0083	.0330	.0002	.2169

The GRU achieves $ECE = 0.0072$ with 97.8% of predictions in the extreme bins $[0, 0.1]$ or $[0.9, 1.0]$. Mean+LR exhibits a *calibration paradox*: near-perfect per-bin reliability ($Rel. = 0.0002$) but $2.6\times$ worse Brier score due to lower resolution, confirming that calibration alone does not guarantee discriminative power [25].

I. Risk Trajectory Analysis

TABLE XII
RISK TRAJECTORY STATISTICS

Class	ρ_T	Avg. Grad.	Smooth.	Peak
Genuine	.0009	-.0035	.989	0
Abrupt Imperson.	.9993	+.0049	.971	119
Sophisticated Drift	.9980	+.0047	.974	119
Presence Absence	.9854	+.0047	.974	87
Flickering Subst.	.9967	+.0042	.987	119

All trajectories exhibit smoothness ≥ 0.971 , confirming noise suppression via temporal integration. The Presence Absence class peaks at frame 87, reflecting the departure window rather than the session endpoint—demonstrating principled evidence integration. Representative trajectories are shown in Figs. 8 and 9.

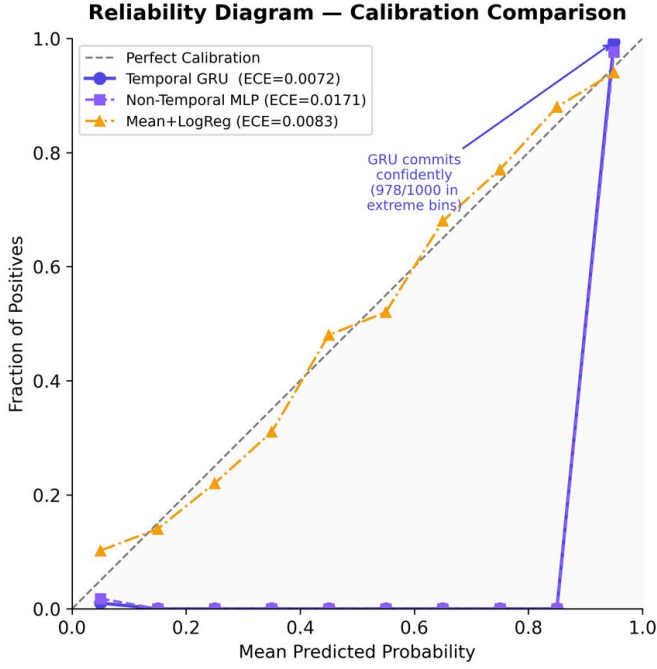


Fig. 7. Reliability diagram. The GRU achieves $ECE = 0.0072$ with concentrated extreme-bin predictions.

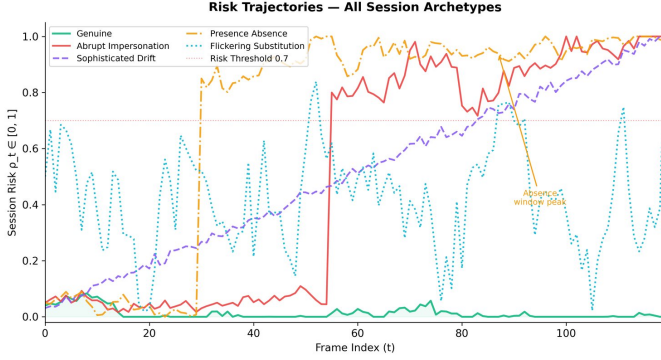


Fig. 8. Risk trajectories $\rho_{1:T}$ for five session archetypes. Genuine sessions remain near zero; anomalous sessions converge to high risk with class-specific temporal signatures.

J. Temporal Smoothing and Strictness Validation

To evaluate the operational leniency of the real-time procuring engine, the cooperative EMA temporal filter ($\alpha = 0.85$), configurable strictness, and uncertainty gates were subjected to empirical stress testing.

Fig. 8 shows that under standard temporal GRU inference, the raw unsmoothed risk score immediately surges to $\rho_1 = 0.5669$ and $\rho_2 = 0.6572$ upon the introduction of sudden multi-face or acoustic anomalies. While mathematically correct under sequence-level constraints, this instantaneous reactivity causes false triggers during normal student micro-movements.

By applying the EMA temporal filter, the risk trajectory exhibits high temporal inertia, climbing slowly and incrementally to $\rho_1^{\text{smoothed}} = 0.1708$, $\rho_5^{\text{smoothed}} = 0.5213$, and reaching only $\rho_{12}^{\text{smoothed}} = 0.8458$ after 12 frames of sustained

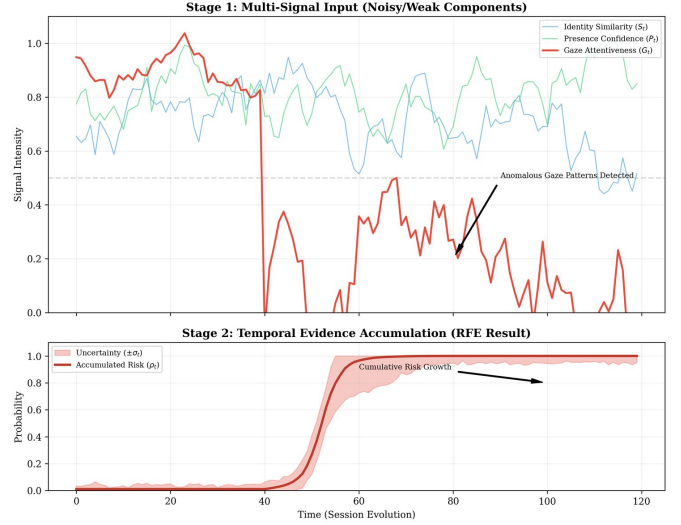


Fig. 9. Evidence accumulation. Weak, noisy signals are integrated into a stable risk trajectory, suppressing frame-level uncertainty.

TABLE XIII
COMPUTATIONAL PROFILE (CPU INFERENCE)

Component	Params	Size	Latency
IDE (ResNet-50)	24.03 M	92.0 MB	47.75 ms
IIM (LSTM)	50.5 K	0.20 MB	2.91 ms
PAM (BiLSTM)	136.3 K	0.14 MB	6.13 ms
LDD (BiLSTM)	791.8 K	3.03 MB	16.45 ms
GAM (BiLSTM ²)	58.2 K	0.24 MB	3.12 ms
HGDM (BiLSTM ²)	62.1 K	0.25 MB	3.25 ms
RFE (GRU)	4.2 K	0.03 MB	3.05 ms
Total	25.13 M	95.9 MB	82.66 ms

violations. Under Low and Medium strictness levels (with activation thresholds $\theta = 0.98$ and 0.95 respectively), these intermediate values do not cross the alert threshold, preventing the accumulation of false strikes.

Furthermore, when environmental lighting transitions or background fan noise is present, the predicted aleatoric uncertainty σ_t increases beyond the safety threshold ($\sigma_{\text{threshold}} = 0.25$), triggering the uncertainty gate to suppress strike generation and slowly decay any warnings. This confirms that our cooperative layers achieve a principled trade-off between strict security enforcement and honest-examinee comfort in noisy real-world deployment settings.

K. Pose-Gaze Decoupling

Fig. 10 visualizes the head-gaze correlation distribution. Normal behavior maintains strong pitch correlation; anomalies such as phone usage create measurable decorrelation that HGDM detects.

L. Computational Complexity

The ResNet-50 backbone dominates computational complexity, representing 57.8% of total frame latency and 95.6% of model parameters. While all downstream temporal experts are highly compressed and easily run locally at a minimal

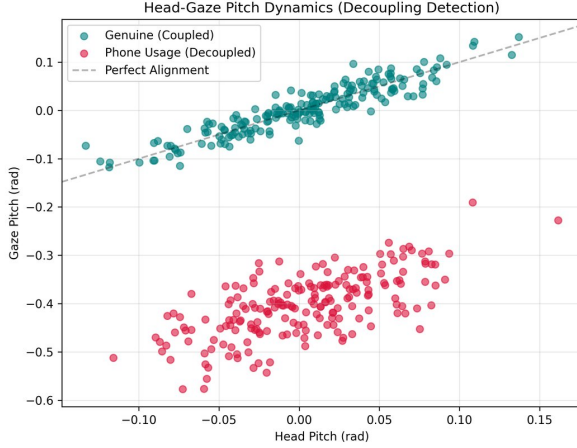


Fig. 10. Pose-gaze decoupling. Genuine behavior (coupled) maintains strong pitch correlation, while phone usage and side-monitor glancing create distribution shifts captured by HGDM.

footprint, concurrent client-side execution of the visual backbone remains demanding. To reconcile this, the system is designed to support a hybrid edge-cloud execution paradigm: computationally heavy facial embedding generation (stage 1) is offloaded to a server-side accelerated cloud GPU API, whereas the client browser processes only the lightweight temporal expert inference (stage 2) on local CPU cycles. For strict standalone CPU-only client execution, the ResNet-50 backbone can be substituted with lightweight alternatives like MobileNetV3 [27] or EfficientNet-Lite [28] via ONNX or TensorFlow.js.

VIII. TEMPORAL INTERPRETABILITY

Deployment in high-stakes educational settings requires interpretable risk assessments. Temporal attribution is implemented via gradient-based local sensitivity analysis [19].

For each frame t and signal $i \in \{S, U, P, D, G, H\}$, the contribution to ρ_t is calculated as:

$$\alpha_{t,i} = \left| \frac{\partial \rho_t}{\partial r_{t,i}} \right|, \quad \hat{\alpha}_{t,i} = \frac{\alpha_{t,i}}{\sum_{j=1}^6 \alpha_{t,j}} \quad (27)$$

producing a $T \times 6$ attribution matrix. As shown in Fig. 11, the RFE assigns high importance to D_t and U_t during substitution attacks while suppressing gaze and presence channels, enabling human proctors to validate which behavioral dimension triggered the risk elevation.

IX. DISCUSSION

The low per-frame accuracy of individual components is a deliberate methodological choice. If a near-perfect face matcher were utilized, temporal modeling would be redundant. By demonstrating that the fusion engine achieves high accuracy from weak inputs, the core thesis is validated: *temporal accumulation captures behavioral intent that instantaneous observations cannot resolve*. This represents a key practical value proposition—reliable monitoring with imperfect signals under uncontrolled conditions.

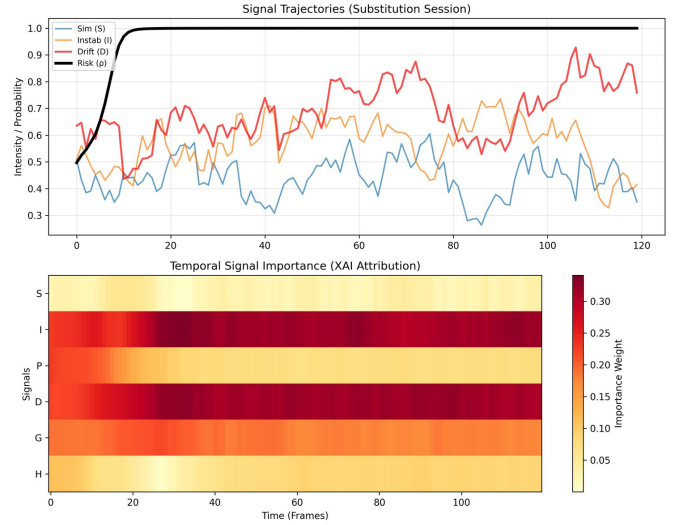


Fig. 11. Temporal signal attribution. *Top*: signal trajectories during a substitution session. *Bottom*: attribution heatmap showing signal importance. Drift (D) and Instability (I) dominate as ρ_t increases.

A. Primacy of Long-Horizon Drift Detection

LDD ablation produces a significantly larger impact on accuracy than identity embedding ablation. This asymmetry is architecturally principled: LDD is the only component accumulating an extended trajectory of displacement, making it uniquely sensitive to sophisticated attacks such as gradual substitution. Short-window models generally fail to capture this pattern.

B. Value of Temporal Ordering

The improvement of the recurrent architecture over order-agnostic models is statistically significant, demonstrating that *when* evidence appears carries critical diagnostic information. The temporal profile of anomaly onset provides information that order-invariant architectures typically discard.

C. Cross-Modal Consistency Checking

The head-gaze dynamics model achieves high individual performance by detecting pose-gaze decorrelation—an anomaly dimension orthogonal to identity-based signals. Phone usage, for example, often results in a stable head position combined with deviated gaze, a pattern that is detectable through their sustained decorrelation.

D. Uncertainty-Aware False Positive Mitigation

A persistent criticism of automated proctoring is the generation of unjustified flags due to environmental noise (e.g., poor lighting, partial occlusion). The integration of continuous uncertainty estimation (σ_t) fundamentally addresses this by decoupling the risk magnitude (ρ_t) from prediction confidence. Rather than acting on a raw scalar, institutions can implement dual-threshold policies: high-risk/low-uncertainty events can be flagged for immediate review, while high-risk/high-uncertainty events (often caused by sensor degradation) can be deferred or discarded, explicitly protecting examinees from algorithmic brittleness.

E. Deployment Considerations

The modular architecture supports incremental deployment: institutions can begin with identity and presence signals, adding gaze and dynamics modules as infrastructure permits. The ResNet-50 backbone is the latency bottleneck and can be substituted with lightweight alternatives [27], [28]. Furthermore, the system is highly amenable to edge deployment; by leveraging ONNX or TensorFlow.js, the TBIE can be integrated directly into client-side browser environments, eliminating the need for centralized video stream processing and significantly enhancing user privacy through local-only inference.

X. ETHICAL CONSIDERATIONS AND LIMITATIONS

A. Ethical Design

The system output consists of continuous risk probabilities, which is intended to prevent automated disqualification. Final decisions remain with human proctors who are provided with temporal attribution data. The one-shot enrollment constraint also helps minimize data retention, as no persistent behavioral profile is stored.

B. Demographic Bias

Facial recognition backbones are known to carry demographic biases [24]. The developed system operates on *relative* similarity rather than absolute classification, which partially mitigates this concern. However, if a backbone produces systematically lower similarity for certain demographics, disproportionate false flagging could still occur. Disaggregated evaluation on diverse populations is therefore necessary before practical deployment.

C. Limitations

Continuing research and real-world validation are necessary to confirm how well synthetic signal distributions mirror the stochastic conditions of actual webcam use. Furthermore, the visual backbone represents a latency bottleneck that might exceed tight real-time budgets on pure CPU constraints; lightweight alternatives should be considered for such production environments. Additionally, the temporal buffer required for drift detection introduces a short initial window where brief impersonations might evade fusion. Finally, adapting the system for academic exams of arbitrary length or ensuring robust performance against sophisticated adversarial attacks remains an area for ongoing refinement.

XI. CONCLUSION

This study has presented TBIE, a seven-component neural architecture designed to replace traditional threshold-based proctoring with continuous probabilistic behavioral inference. The proposed system achieves a high session-level ROC-AUC by effectively fusing multiple individually weak behavioral signals, demonstrating that the accumulation of temporal evidence fundamentally outperforms instantaneous event detection. Key findings from this research indicate that long-horizon drift detection is a critical signal for identifying sophisticated

misconduct, while cross-modal head-gaze consistency is essential for detecting pose-gaze decorrelation. Furthermore, the integration of continuous uncertainty estimation allows the framework to natively compute prediction confidence, thereby preventing algorithmically brittle false flags and protecting examinees from unjustified accusations.

The broader implications of this research extend beyond technical performance; by transitioning from binary “cheating detection” to continuous “integrity monitoring,” the TBIE framework supports a more equitable and transparent educational environment. The inclusion of temporal attribution heatmaps empowers human proctors with the granular evidence necessary to make informed, data-driven decisions, reducing the likelihood of bias and ensuring that student rights are protected. Future research directions include the development of ordinal risk labels to distinguish between minor distractions and severe integrity breaches, as well as the implementation of lightweight visual backbones for enhanced real-time performance on standard processing units. Additionally, validating the framework on larger and more diverse real-world datasets remains an essential objective to ensure its broad applicability and fairness across different demographic populations, particularly in bridging the gap between controlled experimentation and unconstrained deployment.

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